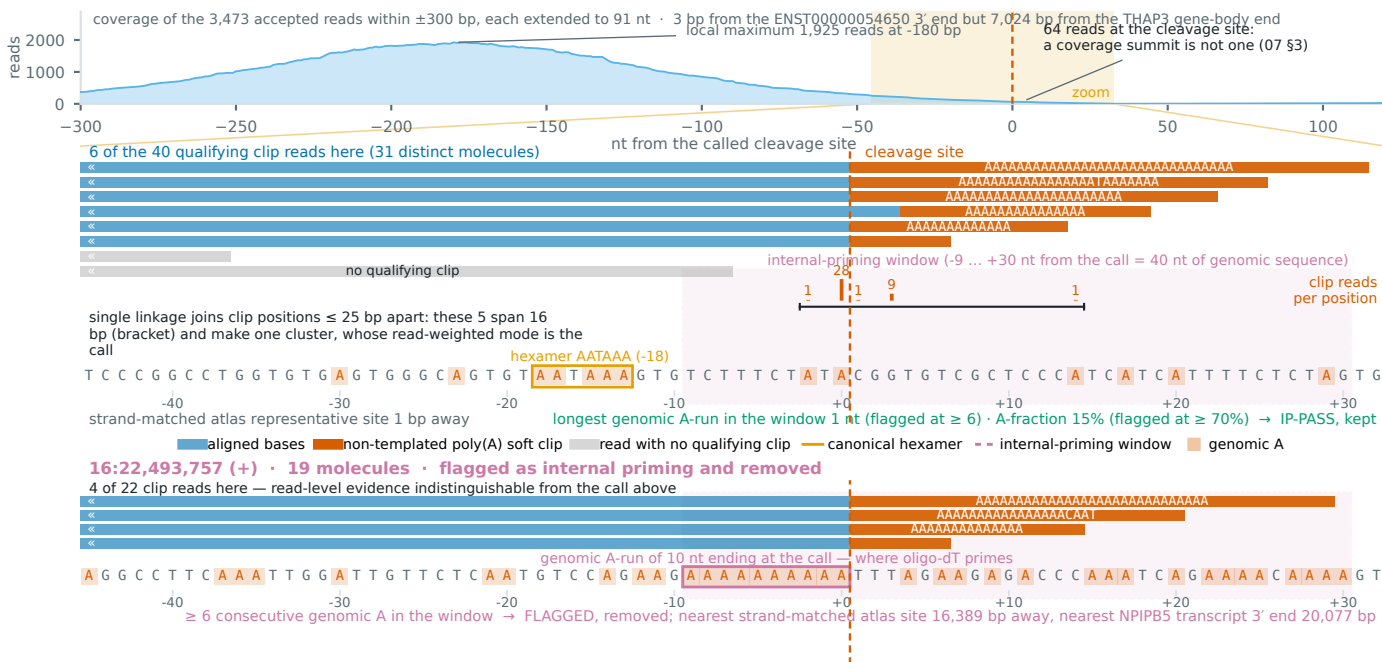


a Read ingestion — what the caller accepts, and the small, highly specific channel it seeds from

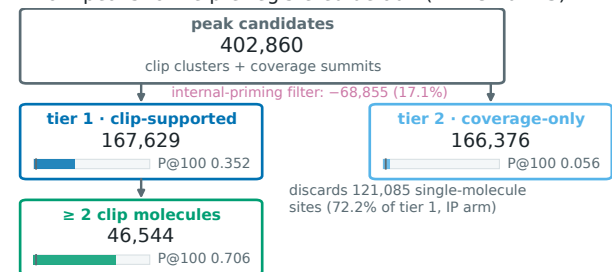


the same usable reads, extended to 91 nt, are the coverage channel (tier 2, panel c) and the count column the switch test uses (panel f): 3,491 reads counted onto the panel-b PAS — the coverage peak it claims plus read ends in [site - 91, site + 25], of which 768 — against 40 clip reads

b One real locus, drawn to scale — 1:6,633,562 (+) · in THAP3 · tier 1, 31 molecules, IP-pass, in the default

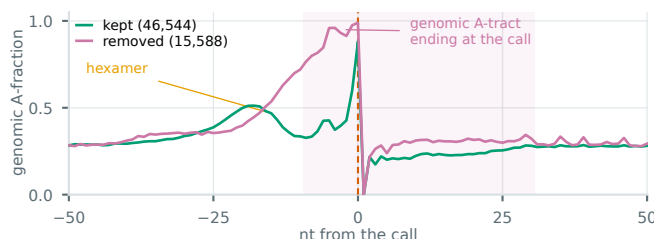


c From peaks to the pre-registered default (PBMC 10k v3)



= the pre-registered default: tier 1 n IP-pass n ≥ 2 distinct clip molecules
gate $P@100 \geq 0.50$: PASS on all three — 0.7062 PBMC, 0.7450 / 0.7572 mice (19 \$1)
3-seed gene-body-shuffled null 0.0217 (grey tick): 33x above chance
tier 2 goes to its own file: not in the default, not switch-tested here
counts are BED rows; each $P@100$ is scored on primary contigs only (n 7-64 lower)

d What the filter removes, measured



15,588 of 62,132 ≥2-molecule calls removed (25.1%); $P@100$ 0.591 → 0.706, R_{det} 0.191 → 0.175

the removed calls score 0.246 against the atlas, the kept ones 0.706 [computed here]

99.97% of removals fire on the ≥6-A run rule, 0.63% on the ≥70% A-fraction rule; the run ends at or before the call in 92%

e One point on a trade surface

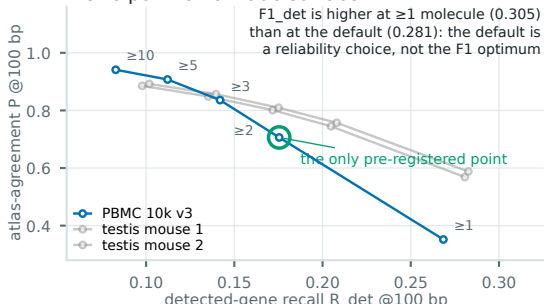


Fig 1 | What PeakATail does, drawn on the data it was measured on (code 9dfdefb; PBMC 10k v3 arm run 2026-08-21 16:14-16:48). (a) A read is used when it is mapped, on the strand of the pass, carries a 16-nt CB tag and aligns over ≤ 91 nt; it enters the poly(A) channel when its 3'-side terminal soft clip is ≥ 6 nt and ≥ 80% A (+) / T (-) with a ≥ 6-nt run flush to the alignment edge — 0.573% of accepted CB reads genome-wide (3,195,067 / 557,564,408; 23 \$2, superseding the 1.152% of 10 \$R2). Support is distinct (CB, UB) molecules. (b) Real BAM records and real GRCh38 sequence; the cluster, its read-weighted mode and its molecule count are recomputed here from those records and reproduce column 5 of the caller's BED. Beneath it, at the same scale, a real ≥2-molecule call the internal-priming filter removed. (c) Sizes and precisions from one scorer (19 \$1/\$2). (d) Profiles, trigger breakdown and the removed set's precision are computed by this script; the same pipeline returns 0.7060 on the kept set, reproducing the scorer's 0.7062. (e) Fig 3's molecule sweep; only ≥2 was pre-registered. (f) Only the FDR-controlling configuration is used, and only replicated switches are reported. Caveats: atlas-agreement precision is agreement with a curated atlas, not ground truth; one donor, one chemistry, and pipelines that trim poly(A) before alignment destroy this evidence (10 \$R2); the panel-b loci are examples chosen by stated criteria, not summaries; --polya-min-umis is 1 at the caller and ≥2 molecules is the pre-registered OUTPUT. Full caption and sources: manuscript/figures/fig1_overview.caption.md; every printed value: results/figures/manuscript/fig1_overview*.tsv.

f What the paper does with the default call set

